

KIMBERLY (KIMI) YASHAR

Cupertino, CA

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EDUCATION

- Carnegie Mellon University, Pittsburgh, PA – Class of 2030
 - Double major: Physics (Applied track) and Entrepreneurship
- The Harker School, San Jose, CA – Class of 2026
 - Relevant Coursework: Neural Networks, Nanoscience, Differential Equations, Multivariable Calculus, AP Physics, Honors Data Structures, Honors Biology, Honors Chemistry
- Test Scores: SAT 1500
- Overall GPA: 4.37/4.8 (weighted)

SKILLS

- Software: Mathematica and Fusion 360 CAD
- AI Tools: InsForge, Nimble, Nebius, HydraDB
- Lab: Chemical etching, microscopy and imaging
- Literature review and scientific writing

RESEARCH AND ENGINEERING EXPERIENCE

- **Materials Science Researcher and Head of Microscopy (May 2025 - May 2026)**
Starostina Lab, Aspiring Scholars Directed Research Program
 - Coauthored copper etchants paper accepted to IEEE MIT Research Technology Conference;
 - Paper title: "[Chemical Etchants for Dihedral Angle Measurements of Twin Boundaries in Pure Copper](#)"
 - Presented findings to 300+ people at Research Expos and Colloquia, including MIT Undergraduate Research Technology Conference (October 2025).
 - Created and taught a 24-student course on microscopy, optical interferometry, and CAD.
 - Tested multiple copper etchants including HNO₃, FeCl₃, and HCl solutions with copper and zinc.
 - Measured interfacial free energy ratio of pure Cu using Keyence Optical Interferometer.
- **Optics Researcher @ University of Cambridge (October 2024 - September 2025)**
Alban-Paccha Lab
 - Analyzed the performance tradeoffs of three optical interconnect modulator technologies (VCSELs, MOS-capacitor, and plasmonic Mach-Zehnder) across data rate, bandwidth, energy-per-bit, and voltage-length product.
 - Selected as one of eight high school students invited to present at the Chinese American Semiconductor Professional Association (CASPA) banquet for this research (September 2025).
- **Biophysics Summer Researcher @ Keck Graduate Institute (June 2024 - July 2024)**
Niemz Lab and Casper Lab
 - Learned next generation sequencing (NGS) procedures and industry style project evaluation methods as the only high school student in two graduate research labs.
 - Designed a procedure to improve genetic screenings for pregnancies;
 - Project title: "Non-Invasive Prenatal Testing for Detecting Congenital Heart Defects Using Next-Generation Sequencing."
 - Performed industry-standard cost and risk evaluation reports for a drug repurposing proposal;
 - Project title: "Evaluating Disulfiram as a Therapeutic Candidate for Lyme-borreliosis: Biological and Economic Insights."
 - Selected from my research groups to present to two committees of KGI investors and professors.
- **Personal Engineering Invention (September 2022 - 2023)**
Friction reducing hairbrush, patents.justia.com/patent/20240138539
 - Engineered a friction reducing hairbrush that forms air bearings around each bristle.
 - Built multiple prototypes, engineering custom circuits and mechanisms to tune its performance.
 - Applied for a patent now pending 2nd Office Action.

LEADERSHIP

- **Founder, Discovery Day (February 2025 - Present)**
[bay-area-discovery-day.vercel.app](#)
 - Founded an annual day of science workshops at Peterson Middle School to counter school budget cuts in the science department.
 - Designed a hands-on curriculum and led science classes for 193 students (40% English 2nd language) featuring chemistry and physics labs.
 - Now training younger members to expand the program to schools who faced similar funding cuts.
- **Founder, Community Tables Cooking Classes (August 2024 - April 2026)**
[community-tables.vercel.app](#)
 - Founded a monthly hands-on cooking class at Sunday Friends, a food distribution and family services center in East San Jose, CA with 1,000+ members.
 - Taught 300+ families in monthly 4 hour classes (in both Spanish and English) and raised \$800 in grant funding from NPO sponsors.
- **Board Member, National Charity League**
President (April 2023 - May 2025), VP of Philanthropy (April 2021 - May 2025)
 - Coordinated philanthropy, culture, and cross-grading bonding events for a chapter of 140+ mothers and daughters, contributing 800+ service hours to shelters, food banks, and education programs.
 - Raised \$27,000 over 5 years for Walk to End Alzheimer's through city-wide fundraising campaigns;
 - Youngest volunteer ever awarded Elite Grand Champion for fundraising efforts.
 - Spoke with 500+ people about Walk to End Alzheimer's at Bay Area community events.

HONORS AND AWARDS

- 1st Prize Winner - Bay Builders Hackathon, AWS Builders Center - July 2026
- National Merit Finalist, Less than 1% of all U.S. high school seniors - 2026
- STEM Merit Scholarship (5% of Applicants)
 - by University of Cambridge, for a year-long optical physics research project
- AP Scholar Award with Distinction - Highest tier of AP Scholar Awards
- Presidential Volunteer Service Award (National): Gold - 2022, 2023, 2024
- California DECA – March, 2025
 - 2nd out of 117 Competitors - Integrated Marketing Campaign
 - 4th out of 94 Competitors - Business Services Marketing Role-play
- Harker DECA – Experienced Leader Award (1 of 189 members)
- \$800 Inspire Innovation Grant Recipient - August, 2024
 - by National Charity League, to one applicant whose initiative is impactful and transformative